

Explainable AI Analysis for Emotion Classification

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Group 11

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Part 1: Gradient $\times$ Input Analysis

Explainable AI (XAI) has gained significant attention as deep learning models are increasingly used in high-stakes applications, where understanding model decisions is crucial for building trust, debugging, and detecting biases. Although transformer-based architectures deliver strong results on emotion classification tasks, their highly complex structures with millions of parameters make it challenging to identify which input features are driving predictions. XAI techniques seek to offer transparency by indicating which segments of the input text the model emphasizes when making classification judgments.

One method for interpreting model behavior is the Gradient $\times$ Input approach. This technique calculates the gradient of the model's output relative to each input token's embedding and multiplies it by the embedding itself. The resulting relevance scores reflect the contribution of each token to the final prediction. Formally, for each token, relevance is computed as the element-wise product of the gradient and the input embedding, summed over the embedding dimensions. Tokens with higher absolute relevance scores are considered to have a stronger influence on the model's classification output.

In this study, we applied the Gradient $\times$ Input method to 18 sentences from our emotion classification dataset, with three sentences representing each of six emotions (anger, disgust, fear, happiness, sadness, surprise). For every sentence, we generated visualizations of the token-level relevance scores and compared the model's highlighted tokens to the emotionally significant words we manually identified. The aim was to assess whether the model attends to semantically meaningful emotion-bearing words or instead relies on irrelevant features such as stopwords or special tokens.

Figure 1 (see Annex at the end of this paper) presents an example of the analysis output, illustrating token-level relevance scores for one of the examined sentences.

Findings

The Gradient $\times$ Input analysis highlighted notable limitations in explaining the decisions of our emotion classification model. Relevance scores were extremely small overall, ranging between 10^{-6} and 10^{-4} , indicating weak gradient signals across the dataset. Despite these low absolute values, relative rankings of tokens revealed some interpretable trends. In total, 13 out of 18 sentences (72%) contained at least one manually identified emotional word among the model's top three most relevant tokens, indicating a partial correspondence between the model's attention and human intuition about emotional content.

Performance differed significantly across emotions. Classifications for surprise, sadness, and disgust were the most accurate, with every sentence in these categories successfully highlighting key emotional words. For instance, in the surprise sentence "I never thought I'd see Pete change" (Annex – Figure 2), the model assigned the highest relevance to "never" (16.50%), "change" (15.65%), and "see" (12.66%), perfectly aligning with the expected emotional content. Similarly, the sadness sentence "We are just making ends meet" (Annex – Figure 3) showed strong alignment, with "making" (14.51%), "meet" (10.30%), and "just" (8.67%) appearing in the top rankings. In contrast, happiness classifications failed entirely; all three sentences were misclassified as neutral, with the model prioritizing proper nouns like "Chef" and "Ramsay" over emotional words such as "thankful" or "proud" (Annex – Figure 4).

A recurring issue across all 18 sentences was the prominence of special tokens. Every sentence had [CLS] or [SEP] tokens appearing among the top five by relevance, often taking the highest positions. In the surprise sentence "This is just unbelievable," [SEP] and [CLS] tokens received 33.49% and 31.99% of relative importance, respectively, while the clearly emotional word "unbelievable" did not appear in the top five (Annex – Figure 5). This pattern suggests that the Gradient $\times$ Input method may primarily reflect the model's reliance on special tokens rather than providing insight into which content words drive emotional understanding.

Part 2: Improved Explanation with Conservative Propagation (Layer-wise Relevance Propagation)

The standard Gradient $\times$ Input approach shows clear limitations when applied to transformer architectures, mainly due to how gradients propagate through multiple layers of attention and normalization. These operations can weaken or distort the relevance signal, resulting in noisy attributions where critical tokens may receive low relevance scores while irrelevant tokens appear highly influential. To mitigate these issues, we employed Conservative Propagation, an enhanced method based on Layer-wise Relevance Propagation (LRP). This technique modifies the backward pass by introducing gradient hooks in the attention layers, which regulate gradient flow and ensure that relevance is preserved as it moves backward through the network. The normalization prevents relevance from disproportionately accumulating in certain layers or being lost during computation. In our setup, we added gradient hooks to all 24 attention layers of the DeBERTa model to adjust gradient computation during the backward pass. The aim was to assess whether Conservative Propagation could more accurately identify emotionally salient tokens compared to the basic gradient-based method, reducing the impact of special tokens and irrelevant words while emphasizing genuine emotional content.

Findings

The Conservative Propagation implementation yielded results that were largely unexpected. Contrary to our hopes, this modified method showed almost no improvement over the standard Gradient $\times$ Input approach. Of the 18 sentences analyzed, 17 (94.4%) performed identically across both methods in terms of identifying key emotional words within the top five ranked tokens. In one case (Annex – Figure 6: sadness_o3), Conservative Propagation actually performed worse, failing to capture a keyword that Gradient $\times$ Input had successfully highlighted. No sentences demonstrated improvement with the LRP-based technique. This near-identical performance suggests that, although gradient hooks were applied to all 24 attention layers, they did not substantially change how relevance was distributed across tokens for this specific model and dataset.

Magnitude analysis showed that Conservative Propagation produced considerably smaller relevance scores overall. The average relevance magnitude dropped by 66% (from 1.35e-05 to 4.65e-06), and the maximum scores decreased by 77% (from 1.21e-04 to 2.79e-05). Despite these reductions, the relative token rankings remained mostly unchanged, explaining why both methods highlighted the same key words. The gradient hook normalization compressed the score range but preserved the relative ordering of importance. This indicates that while Conservative Propagation effectively modified gradient flow as intended, it did not fundamentally alter which tokens the model deemed most relevant for predictions.

The issue of special token dominance persisted under both methods. [CLS] and [SEP] tokens frequently appeared among the top-ranked positions in multiple sentences, and gradient normalization in Conservative Propagation did not alleviate this. For example, in surprise_o3, both methods ranked special tokens highest while the emotionally salient word “unbelievable” failed to appear in the top five. Similarly, for happiness sentences misclassified as neutral, both methods focused on proper nouns like “Chef” and “Ramsay” rather than emotional words such as “thankful” or “proud.” These results suggest that the prevalence of special tokens may be intrinsic to how the model represents and processes information, rather than an artifact of the attribution method.

Part 3: Input Perturbation Analysis

Input perturbation analysis assesses model robustness by systematically removing tokens and monitoring changes in prediction confidence. This approach helps determine whether a model relies on a small set of crucial tokens or incorporates information spread across the entire input. For this study, we used

relevance scores from Conservative Propagation to guide the token removal order, eliminating tokens from least to most relevant while tracking the model's confidence in its original prediction at each step. For each of the 18 sentences, content tokens were removed one by one, and confidence levels were recorded after each removal, producing line graphs that visualize confidence degradation curves. A steep drop in confidence after removing only a few tokens indicates heavy reliance on specific words, signaling brittleness and potential susceptibility to adversarial inputs. In contrast, a gradual decline in confidence suggests that the model distributes information across multiple tokens, reflecting more robust and stable representations. This analysis sheds light on which emotion categories the model handles more reliably and which predictions may be less trustworthy despite initially high confidence scores.

Findings

The perturbation analysis uncovered clear differences in robustness across the 18 sentences. Ten sentences (55.6%) exhibited sharp confidence drops, losing 50% of their prediction confidence after removing only 1–6 tokens, while eight sentences (44.4%) showed more gradual declines, reflecting distributed token importance. The most dramatic drop was observed in `disgust_02` (“That's disgusting”), where confidence fell by 33.3% per token over the first three removals, dropping below 50% after just two tokens despite starting at 99.8% confidence (Annex – Figure 7). This indicates that the model's disgust classification relied almost entirely on the single word “disgusting,” making it highly sensitive to removal or replacement of that token. In contrast, `anger_03` (“It's f*cking frustrating”) demonstrated notable robustness, with a negative drop rate of –5.8% per token; confidence either stabilized or slightly increased as irrelevant tokens were removed (Annex – Figure 8). This counterintuitive outcome suggests that eliminating noise improved the model's focus on core emotional content.

Robustness varied substantially by emotion category. Happiness predictions were the most stable, maintaining an average confidence of 78.9% across perturbations, with all three happiness sentences showing gradual degradation. For instance, `happiness_02` retained over 50% confidence even after removing 12 tokens, indicating that the model distributed attention across multiple words. Surprise classifications were also robust, averaging 72.2% confidence, with `surprise_01` never falling below 50% throughout the perturbation sequence. This suggests that the model captures surprise through several linguistic cues rather than depending on individual keywords. Conversely, disgust was the least robust, averaging only 37.8% confidence, with all three disgust sentences showing steep drops. `Disgust_01` (“It looks like a giant diaper stuffed with meatballs”) lost 50% confidence after removing just two of its ten content tokens.

Sentence length alone did not predict robustness. Although `disgust_02` contained only three content tokens and showed sharp drops, `fear_01` (“We got into trouble”), with four content tokens, maintained a gradual decline, requiring all four tokens to be removed before confidence fell below 50%. This indicates that robustness depends more on how the model distributes attention across tokens rather than sentence length. Anger sentences exhibited mixed patterns: two showed gradual confidence declines, while one experienced a sharp drop, resulting in moderate overall robustness (55.6%). The `anger_01` sentence (“Why don't you get the f*ck out of my restaurant?”) retained confidence through six token removals, suggesting the model successfully integrated multiple aggressive terms rather than relying solely on profanity.

Interestingly, several sentences misclassified in earlier analyses displayed relatively robust behavior. All three happiness sentences incorrectly labeled as neutral exhibited gradual confidence drops, indicating the model was consistently confident in its wrong predictions. This underscores a key point: robustness measures the distribution of evidence across tokens, not prediction accuracy. A model can be confidently robust while still making incorrect predictions if it consistently misinterprets distributed cues. The perturbation analysis also highlighted potential critical keywords for each emotion. For anger, removing “f*ck” or “frustrating” caused major confidence declines; for fear, “trouble” and “lose” were pivotal; and for sadness, “customers” and “sell” proved essential. While these findings align with human intuition about emotional language, the reliance on single tokens in some cases—especially for disgust—suggests the model may not fully capture contextual nuances that distributed patterns would reveal.

Summary and Conclusions

This explainability analysis of our emotion classification model for the Content Intelligence Agency provided key insights into how the system identifies emotions in transcribed video content. Using the Gradient $\times$ Input method, the model successfully highlighted emotional keywords in 72% of sentences, but the attribution signals were generally weak. Conservative Propagation offered little improvement, producing nearly identical token rankings in 94.4% of cases. The perturbation analysis yielded the most informative results: 55.6% of predictions depended heavily on just 1–3 tokens, with robustness varying across emotions. Happiness (78.9%) and surprise (72.2%) predictions were more distributed and reliable, while disgust predictions (37.8%) proved fragile and highly keyword-dependent.

Despite these limitations, the model delivers significant value for automated emotion tagging within the Content Intelligence Agency’s workflow. Predictions for surprise (72.2%), fear (60.4%), and anger (55.6%) are robust enough to be production-ready. The system enables content creators to search video libraries by emotional content, generate emotional timelines, and extract highlights for trailers, all while processing thousands of hours at a fraction of the cost of manual annotation.

Overall, this study showed that while the model handles several emotions effectively with distributed attention, it struggles with happiness and exhibits brittleness in disgust classification. Gradient-based interpretability methods provided only limited insights, whereas perturbation analysis offered the clearest understanding of robustness. For deployment, the model can be used with appropriate confidence thresholds and human review for sensitive or problematic categories. Future improvements—such as incorporating domain-specific data, expanding contextual understanding, and adding multi-modal features—would enhance reliability for large-scale media content analysis.

Future Improvements

Several enhancements could further improve model performance for the Content Intelligence Agency’s pipeline. Domain-specific training using film and TV transcriptions would better capture scripted dialogue patterns, while synonym augmentation could reduce reliance on single keywords by teaching the model that words like “disgusting,” “revolting,” and “gross” convey similar emotions. Temporal smoothing that considers surrounding predictions would produce more stable emotion tracks, reflecting the persistence of emotions over time in video content. Multi-modal integration—incorporating audio cues, such as shouting for anger detection, and visual signals, like facial expressions for happiness—would improve classification accuracy and reduce misclassifications.

In addition, implementing emotion-specific confidence thresholds across all categories—not just happiness and disgust—would allow the system to automatically flag low-confidence predictions for human review. These flagged cases could then be added to future training datasets, helping the model learn from its mistakes and improve performance over time. The current system already allows content creators to search libraries by emotional content, generate timelines, and identify highlights, analyzing thousands of hours cost-effectively. With the addition of confidence-based human review and feedback loops, the model would be even more reliable and ready for deployment in the agency’s media analysis workflow.

Annex

Analysis Table for Gradient $\times$ Input Results – Figure 1

Sentence ID	Sentence Text	Your Key Emotional Words	Model's Top 3 Tokens (excluding special)	Match?	Special Tokens in Top 5?	Assessment
anger_01	Why don't you get the fuck out of my restaurant?	fuck, get out	restaurant, my, fuck	Partial	No	"fuck" ranks 5th (9.46%) but "restaurant" and "my" rank higher - questionable focus
anger_02	What the heck are you laughing about?	laughing about, what the heck	What, about, the	Partial	Yes ([SEP])	"What" and "about" captured, but "laughing" missing from top 5
anger_03	It's fucking frustrating.	frustrating, fucking	fucking, It, 's	Good	Yes ([CLS], [SEP])	"fucking" ranks 2nd (22.97%) - correctly identified, but special tokens dominate
disgust_01	It looks like a giant diaper stuffed with meatballs.	diaper, stuffed	diaper, with, .	Good	Yes ([SEP], [CLS])	MISCLASSIFIED as surprise - "diaper" correctly highest (23.76%), but wrong prediction
disgust_02	That's disgusting.	disgusting	That, disgusting, 's	Good	Yes ([CLS], [SEP])	"disgusting" ranks 3rd (16.20%) - identified but weaker than expected
disgust_03	It's like eating a patch of soaking wet grass	eating, soaking wet grass	wet, soaking, of	Good	Yes ([SEP])	"wet" (25.57%) and "soaking" (8.48%) captured - reasonable focus
fear_01	We got into trouble.	trouble	trouble, into, got	Excellent	Yes ([SEP])	"trouble" dominates (32.65%) - best alignment so far
fear_02	don't want to lose my home and the few little things	lose, home, few little things	the, little, lose	Partial	Yes ([CLS], [SEP])	"lose" only 4th (9.29%), stopwords "the" and "little" rank higher
fear_03	We're going to have to find jobs.	have to, find jobs	have, to, jobs	Good	Yes ([CLS], [SEP])	MISCLASSIFIED as sadness - "have" (18.63%), "jobs" (7.20%) captured
happiness_01	I am so thankful to Chef Ramsay to bring me over here	thankful	Chef, over, Ramsay	Poor	Yes ([SEP], [CLS])	MISCLASSIFIED as neutral - "thankful" NOT in top 5, focuses on proper nouns instead
happiness_02	I am very proud of my meatballs, and for Chef Ramsay	very proud, my meatballs	meat, very, Ramsay	Partial	Yes ([CLS])	MISCLASSIFIED as neutral - "very" captured (9.74%), "proud" missing entirely
happiness_03	I have a beautiful restaurant now.	beautiful	restaurant, beautiful, now	Good	Yes ([SEP], [CLS])	MISCLASSIFIED as neutral - "beautiful" ranks 3rd (14.05%) but wrong emotion predicted

sadness_01	Now, we have no customers.	no customers	Now, customers, .	Good	Yes ([SEP], [CLS])	"customers" ranks 4th (10.43%), "Now" also high - reasonable
sadness_02	We are just making ends meet.	just, making ends meet	making. We, meet, just	Excellent	Yes ([SEP])	"making" (14.51%), "meet" (10.30%), "just" (8.67%) all captured - good alignment
sadness_03	we're going to have to sell our home.	sell, home	sell, going, to	Good	Yes ([SEP])	"sell" ranks 3rd (13.45%) - identified correctly
surprise_01	Wow, and how do you, how would you eat that?	wow, how	Wow, how, how	Excellent	Yes ([CLS])	"Wow" (21.04%) and "how" (16.89%, 12.80%) dominate - perfect alignment
surprise_02	I never thought I'd see Pete change.	never thought, see...change	never, change, see	Excellent	Yes ([CLS], [SEP])	All key words captured: "never" (16.50%), "change" (15.65%), "see" (12.66%)
surprise_03	This is just unbelievable.	unbelievable	just, is	Poor	Yes ([SEP], [CLS])	"unbelievable" NOT in top 5 - special tokens dominate (33.49%, 31.99%)

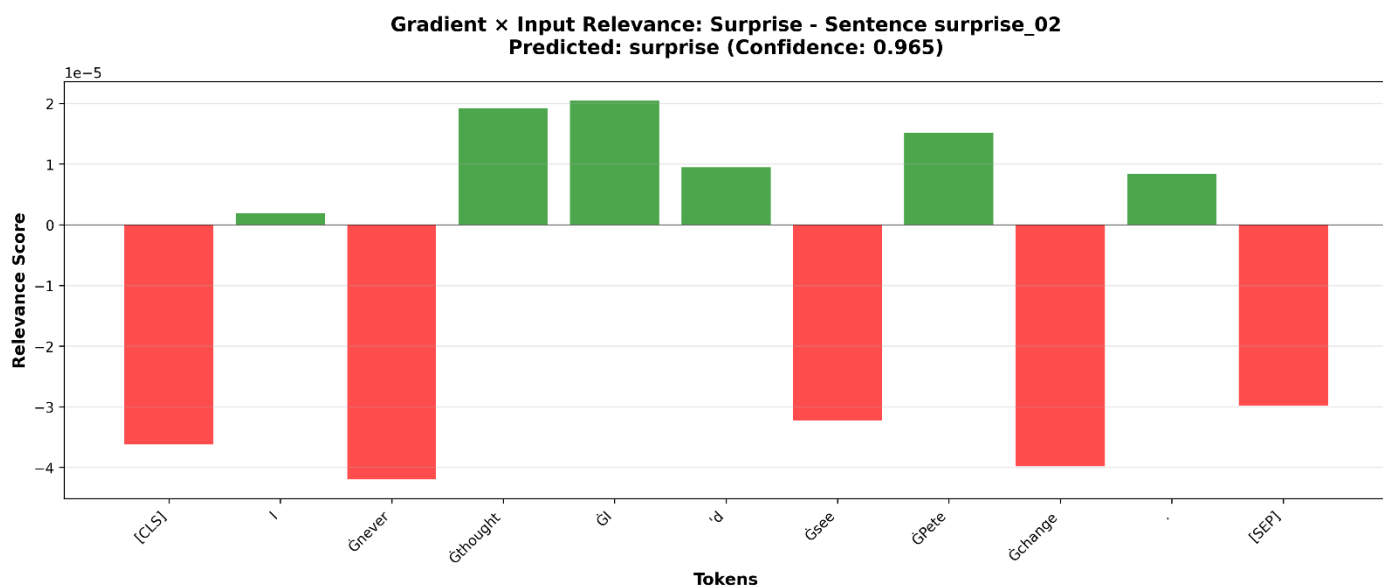


Figure 2

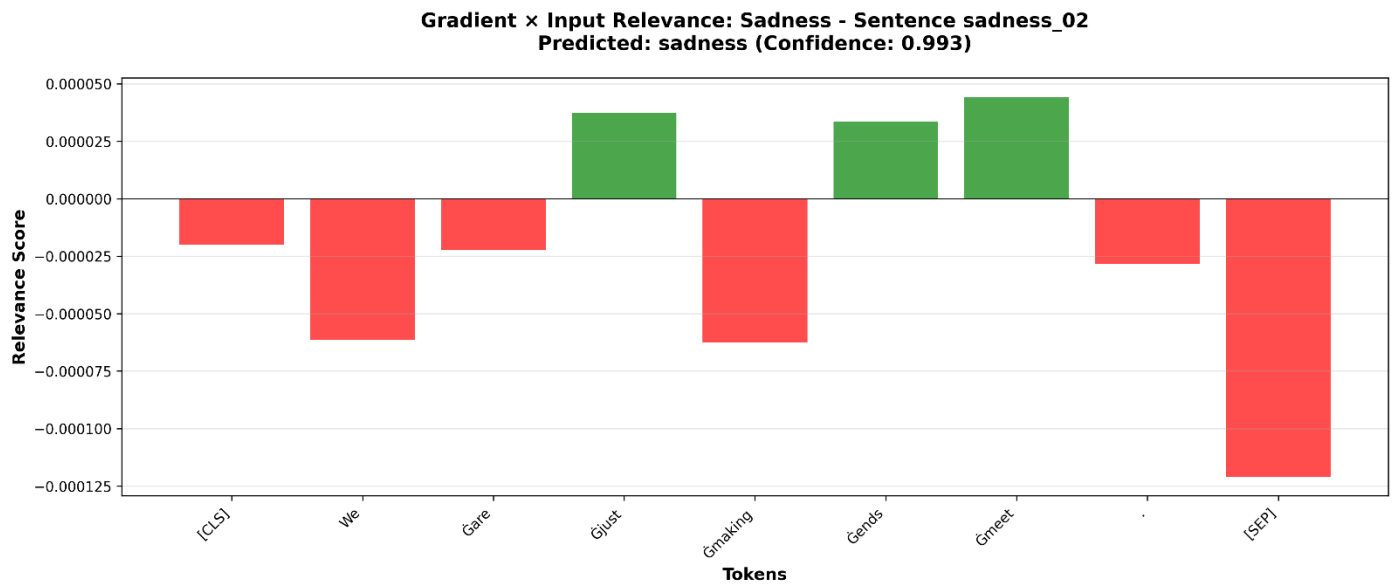


Figure 3

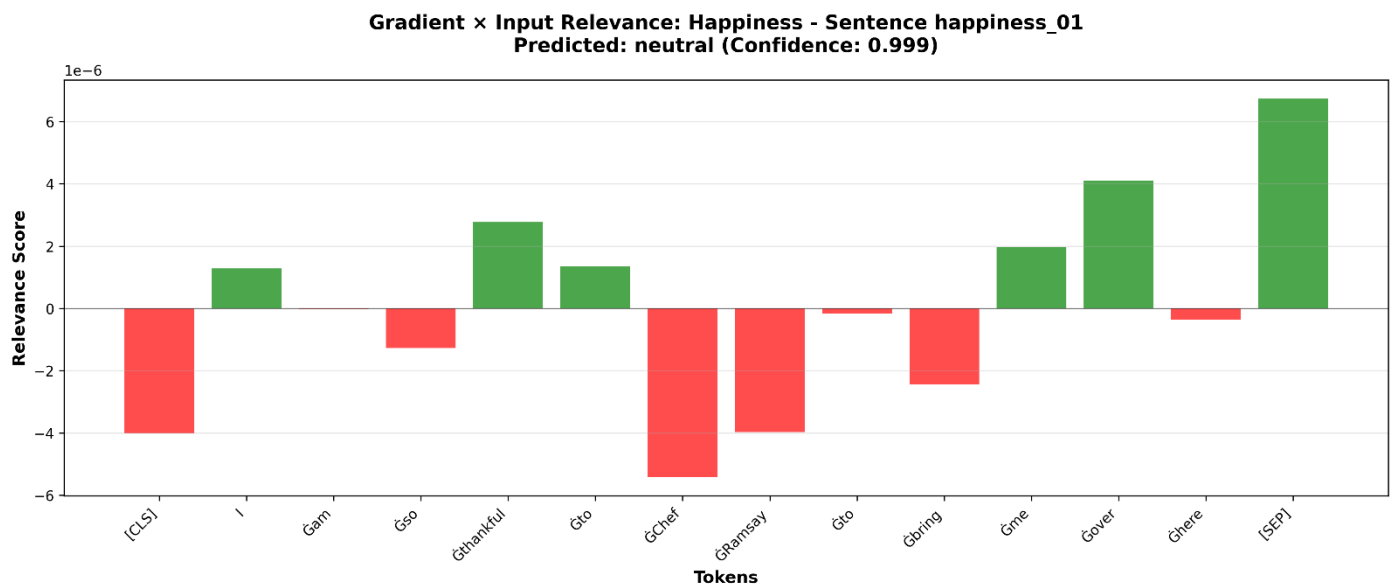


Figure 4

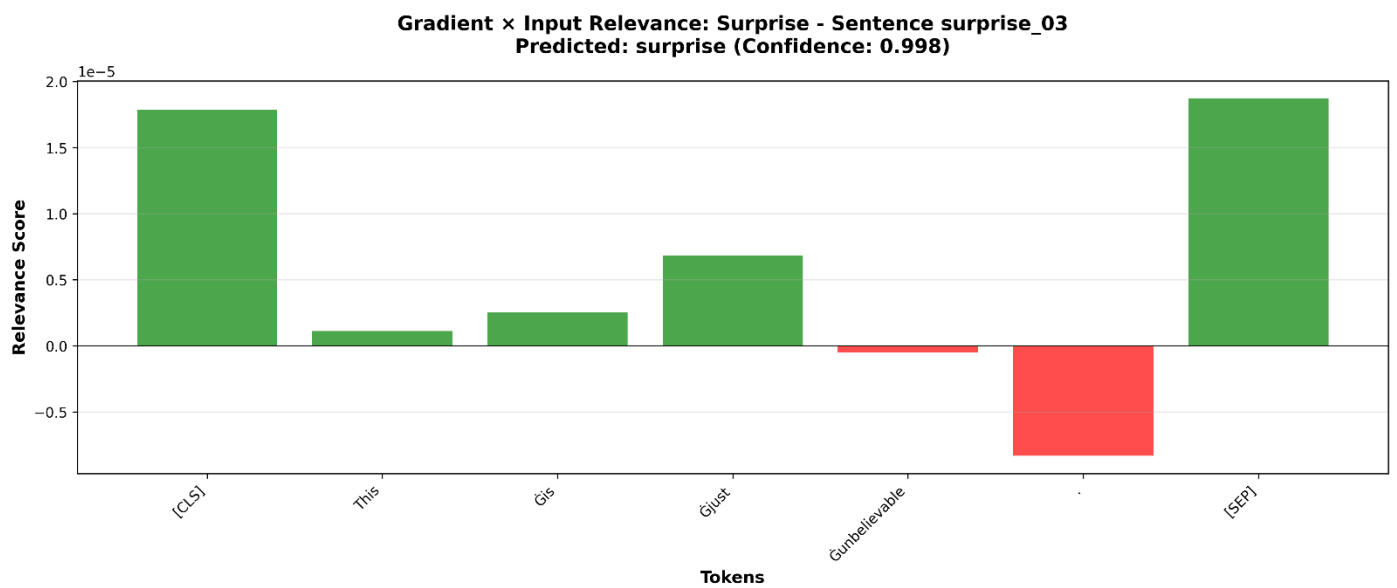


Figure 5

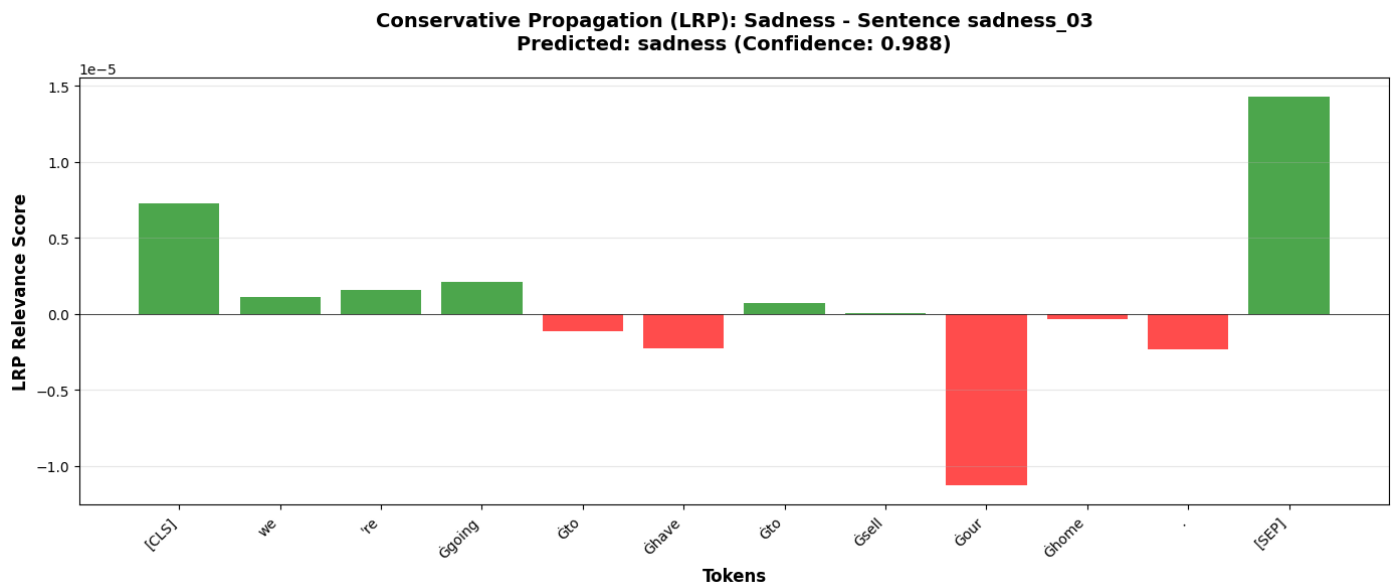


Figure 6

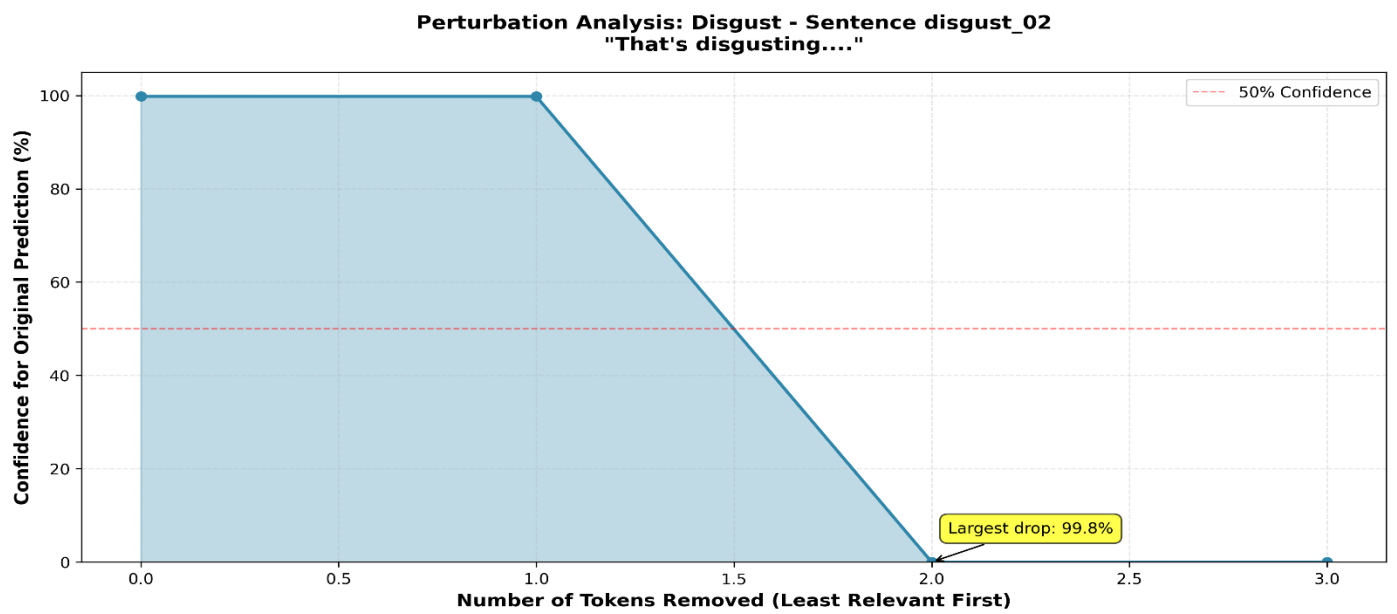


Figure 7

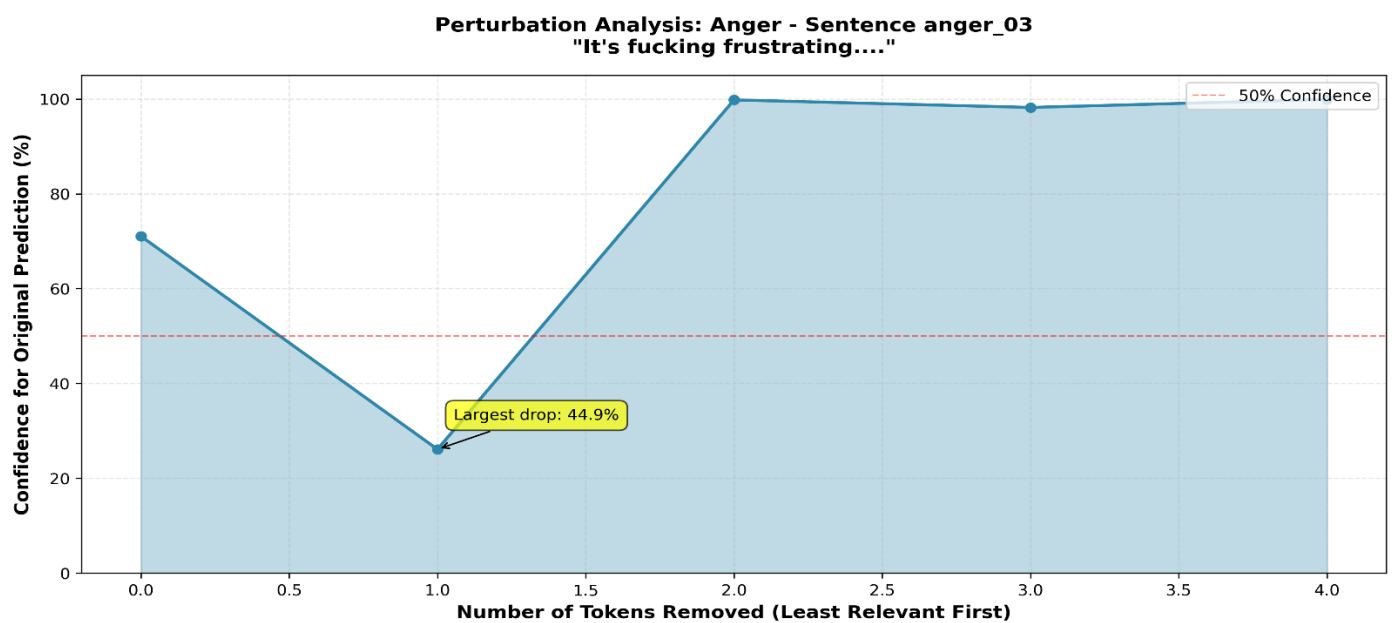


Figure 8

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